

Given condition :

$$\text{coal production per year} = 4.5 \times 10^6 \text{ ton}$$

$$\text{stripping ratio} = 2.5 \text{ m}^3/\text{ton}$$

$$\text{seam thickness} = 10 \text{ m}$$

Assumptions :

$$\text{Bank density of coal seam} = 1.4 \text{ ton/m}^3$$

$$\text{Bank density of OB} = 2.6 \text{ ton/m}^3$$

$$\text{Bucket capacity of coal} = 5 \text{ m}^3$$

$$\text{Bucket capacity of OB} = 5 \text{ m}^3$$

$$\text{Bucket fill-factor of coal} = 0.85$$

$$\text{Bucket fill-factor of OB} = 0.80$$

$$\text{Swell factor (coal + OB)} = 20 \%$$

$$\text{Swing factor (90°)} = 1$$

$$\text{Shovel cycle time (coal + OB)} = 30 \text{ s}$$

$$\text{Rated capacity of dumper (both coal \& OB)} = 50 \text{ ton}$$

$$\text{Total working hrs / year} = 300 \times 3 \times 8 = 7200 \text{ hrs}$$

$$\text{Spotting time} = 0.3 \text{ min}$$

$$\text{Dumping time} = 1.5 \text{ min}$$

$$\text{Load travel speed of dumper} = 20 \text{ km/hr}$$

$$\text{Empty travel speed of dumper} = 30 \text{ km/hr}$$

$$\text{Haul distance (one way for both coal \& OB)} = 1 \text{ km}$$

$$\text{Overall efficiency for 3 shifts} = 0.56$$

$$\begin{aligned} \text{Average OB thickness} &= \text{S.R.} \times \text{Coal seam thickness} \times f_{\text{coal}} \\ &= 2.5 \times 10 \times 1.4 \\ &= 35 \text{ m} \end{aligned}$$

Three benches with bench height 12m, 12m & 11m.

★ Calculation for no. of shovel for coal : —

$$\text{Production per year} = 4.5 \times 10^6 \text{ ton}$$

$$Q_{eff} = 5 \times 0.85 \times \frac{100}{(100+20)} \times 7200 \times \frac{3600}{30 \times 1} \times 0.56 \times 1.4 \text{ ton}$$

$$= 2399040 \text{ ton}$$

$$\text{Number of working shovels req.} = \frac{4.5 \times 10^6}{2399040}$$

$$= 1.875$$

$$\text{Actual no. of working shovels req.} = 2$$

★ Calculation for no. of overburden shovels —

$$\text{Req. excavation capacity per year} = S.R. \times \text{production of coal}$$

$$= 2.5 \times 4.5 \times 10^6$$

$$= 11.25 \times 10^6 \text{ m}^3$$

$$\text{Effective excavation capacity per year per shovel}$$

$$= 5 \times 0.8 \times \frac{100}{(100+20)} \times 7200 \times \frac{3600}{30 \times 1} \times 0.56$$

$$= 1612800$$

$$\text{No. of shovels req.} = \frac{11.25 \times 10^6}{1612800}$$

$$= 6.975$$

$$\text{Actual no. of shovels req.} = 7$$

★ Calculation of no. of dumpers for coal : —

$$\text{Effective dumper capacity} = 50 \times \frac{1.4}{2.6} = 27 \text{ ton}$$

$$\text{Tonnes per pass of shovel bucket} = B_c \times B_f \times S_f \times \rho_{\text{coal}}$$

$$= 5 \times 0.85 \times \frac{100}{(100+20)} \times 1.4$$

$$= 4.958$$

$$\text{No. of passes req. to load the dumper} = \frac{27}{4.958} = 5.44$$

$$\text{Actual no. of passes} = 5$$

$$\begin{aligned}\text{Dumper loading time} &= \text{Actual no. of passes} \times \text{shovel cycle time} \\ &= 5 \times 30 \\ &= 2.5 \text{ min}\end{aligned}$$

$$\text{spotting time} = 0.3 \text{ min}$$

$$\text{load travel time} = \frac{1}{20} \times 60 = 3 \text{ min}$$

$$\text{Empty travel time} = \frac{1}{30} \times 60 = 2 \text{ min}$$

$$\text{Dumping time} = 1.5 \text{ min}$$

$$\text{Total time} = 9.3 \text{ min}$$

$$\text{Total no. of dumpers req.} = \frac{9.3}{0.3 + 2.5} = 3.32$$

$$\text{Actual no. of dumpers} = 4$$

$$\begin{aligned}\text{waiting time of each dumper} &= (4 \times 2.8) - 9.3 \\ &= 1.9 \text{ min}\end{aligned}$$

$$\text{Total no. of dumpers for 2 shovel} = 8$$

★ Calculation of no. of dumpers for overburden —————

$$\text{Rated dumper capacity} = 50 \text{ ton}$$

$$\begin{aligned}\text{Tonnes per pass shovel} &= B_f \times B_c \times S_f \times P_{OB} \\ &= 5 \times 0.80 \times \frac{100}{(100 + 20)} \times 2.6 \\ &= 8.66\end{aligned}$$

$$\text{Req. no. of passes to load dumper} = \frac{50}{8.66} = 5.77$$

$$\text{Actual no. of passes} = 5$$

$$\text{Dumper loading time} = 5 \times 30 \text{ s} = 2.5 \text{ min}$$

spotting time = 0.3 min

Load travel time = 3 min

Dumping time = 1.5 min

Empty travel time = 2 min

Total time = 9.8 min

Req. no. of dumpers per shovel = $\frac{9.8}{2.8} = 3.52$

Actual no. of dumpers per shovel = 4

Total no. of dumpers = $4 \times 7 = 28$